Model Training Results

Detection and Attribution of AI-Generated and Human Written Phishing Emails from Legitimate Emails

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1 Overview

This is the first run of the full pipeline end to end: real feature extraction (not proxy features), real statistical gating on the extracted values, parameter locking, and evaluation on the independent holdout set. Two findings from this run contradict earlier, proxy-feature-based claims and are reported here rather than quietly corrected.

2 Feature Extraction – what was actually used

All 20 features were computed with real tooling, not approximations: spaCy (`en_core_web_sm`) for dependency parsing and POS tagging, a from-scratch trigram language model (stupid backoff) trained once on the training set's own text for token-predictability features, VADER for sentence-level sentiment, and `pyspellchecker` for orthographic error detection. A pretrained transformer language model was considered for the token log-probability feature and rejected: it was unreachable from this environment (no route to download pretrained weights), and a heavy black-box model would also cut against the project's own stated design goal of a lightweight, fully auditable pipeline. Email text is capped at 6,000 characters before parser/tokenizer-based features run, since a handful of extreme outliers (up to 31,396 words, almost certainly mbox-thread artifacts) would otherwise dominate runtime for no signal benefit.

3 Statistical Gating – real results

Kruskal-Wallis, Mann-Whitney U, Chi-Square, and joint Benjamini-Hochberg FDR correction were re-run on the real extracted values (Section 3 methodology, unchanged). Table 1 summarizes the outcome.

All 20 of 20 features clear the main gate; **16 of 20 clear the attribution gate** (urgency_density, pronoun_balance, bracketed_cta_flag, and threat_clause_flag do not – each is statistically significant purely from sample size, but too small an effect to usefully separate Human-Phishing from AI-Phishing).

A claim this contradicts

`proposal.tex` currently describes imperative verb density as "the single strongest attribution signal in our feature set." Measured against real extracted data, it is not: its attribution effect size

	Main (3-group)			Attribution (2-group)		
Feature	adj. p	effect	gate	adj. p	effect	gate
avg_token_logprob	≈ 0	0.956	✓	≈ 0	−0.956	✓
sentence_perplexity_variance	≈ 0	1.251	✓	≈ 0	0.835	✓
sentence_length_std	≈ 0	0.630	✓	≈ 0	0.263	✓
max_parse_depth	≈ 0	0.637	✓	≈ 0	−0.637	✓
paragraph_density	≈ 0	3.150	✓	≈ 0	−3.150	✓
clause_density	≈ 0	0.405	✓	≈ 0	−0.405	✓
orthographic_error_density	≈ 0	1.431	✓	0.0099	0.406	✓
punctuation_entropy	≈ 0	0.889	✓	≈ 0	−0.541	✓
urgency_density	≈ 0	1.325	✓	≈ 0	−0.042	fails
pronoun_balance	≈ 0	0.992	✓	≈ 0	−0.044	fails
sentiment_variance	≈ 0	0.713	✓	≈ 0	−0.631	✓
imperative_density	≈ 0	0.225	✓	≈ 0	0.225	✓
lexical_diversity	≈ 0	0.514	✓	≈ 0	−0.395	✓
pronoun_density	≈ 0	0.969	✓	≈ 0	−0.354	✓
generic_greeting_flag	≈ 0	0.411	✓	≈ 0	0.240	✓
subject_refw_flag	≈ 0	0.511	✓	≈ 0	0.158	✓
attachment_mention_flag	≈ 0	0.168	✓	≈ 0	0.120	✓
bracketed_cta_flag	≈ 0	0.377	✓	0.0004	0.065	fails
sensitive_info_flag	≈ 0	0.306	✓	≈ 0	0.175	✓
threat_clause_flag	≈ 0	0.168	✓	0.0055	0.051	fails

Table 1: Dual-gate results on real extracted features (adjusted $p < 0.05$ **and** effect size at or above threshold: $|d| \geq 0.20$ continuous, $V \geq 0.10$ categorical). Effect-size sign for the attribution column indicates direction (which class scores higher), not magnitude – the gate itself is applied to the absolute value.

($d = 0.225$) is one of the smallest that still clears the gate, while `paragraph_density` ($d = 3.15$), `avg_token_logprob` ($d = 0.96$), and `sentence_perplexity_variance` ($d = 0.84$) are all substantially larger. The proposal's claim was based on an earlier qualitative expectation, not this measurement, and should be corrected before the paper is finalized.

A result worth treating with suspicion, not celebration

`paragraph_density`'s effect size ($d = 3.15$) is far larger than any other feature's – large enough to be a red flag rather than good news. The three classes come from three different raw sources with three different formatting conventions (Nazario phishing text, raw Enron plain-text email, and LLM-generated output), so a feature this strongly separated is more likely picking up a corpus-formatting artifact (how each source happens to insert paragraph breaks) than a genuine linguistic signal that would generalize to phishing emails collected a different way. This is the same category of caveat already on record for other features (the corpus-era markers in the Legitimate class, the AI-Phishing corpus's 0% exclamation marks). It is flagged here rather than presented as a strong result.

4 Parameter Locking – two pipelines, not one

A design decision was required that the proposal's equations did not fully pin down: whether Step 2 (attribution) reuses the same 3-class posterior as Step 1, or is a separately trained model restricted to phishing-only data and the attribution-gate feature subset. The abstract's own wording – "a dedicated attribution-specific feature subset" – only makes sense as a separate model, since several main-gate features (urgency_density, pronoun_balance) do not actually hold up for attribution. **Two pipelines were therefore built**: a 3-class *main pipeline* (Legitimate / Human-Phishing / AI-Phishing, 20 features) for Step 1, and a 2-class *attribution pipeline* (Human-Phishing / AI-Phishing, trained only on phishing rows, 16 features) for Step 2.

Two other open parameters were fixed: **8 quantile bins** for the Rank-Binned NB (a middle ground for roughly 1,497 training rows per class), and the Meta-NB combination was implemented as $P_{\text{meta}}(C_k) \propto P_1(C_k) \cdot P_2(C_k) \,/\, P(C_k)$, renormalized over k – the self-consistent form of Eq. 9 once $P(P_m \mid C_k)$ is read as $P_m(C_k)/P(C_k)$, which avoids triple-counting the class prior (each sub-model's own posterior already carries one copy of it).

5 Evaluation on the Holdout Set

This is the real generalization test: the holdout set's Legitimate rows are overwhelmingly raw Enron email (4,934 of 4,976), not the Nazario-legit mix the model trained on – deliberately the harder test. Human-Phishing holdout rows are few (42), a direct consequence of the earlier balancing step, so that side of the attribution table is noisier than the AI-Phishing side.

Stage	**Accuracy**	**Precision**	**Recall**
Step 1 (Legitimate vs. Phishing)	0.854	0.776 (Phishing)	0.915 (Phishing)
Step 2 (Attribution, true-phishing rows only)	0.865	–	0.952 (Human-Phishing, n=42) / 0.863 (AI-Phishing, n=3489)
End-to-end 3-class	0.817	–	–

Table 2: Holdout evaluation. Step 2 is scored by running the attribution model directly on every true-phishing row, independent of whether Step 1 actually flagged that row as phishing, so Step 1's errors cannot contaminate this number.

Step 1's false-positive rate on Legitimate email is 18.9% (934 of 4,934 raw Enron emails flagged as phishing) – notably worse than the 99.4% accuracy of the earlier TF-IDF + Logistic Regression sanity baseline from the data-validation stage. This gap is expected and informative, not a bug: the baseline sees the full vocabulary of every email, while this pipeline deliberately restricts itself to 20 interpretable statistical features. The gap is the real, measured cost of that interpretability constraint, and should be reported as such in the paper rather than compared directly against the baseline's number without this context.

6 Open Items

1. `proposal.tex` needs correcting on the imperative-density claim above, and updating from 14 to 20 features (Section 2.1), which was already outstanding.

2. The large `paragraph_density` effect size should be investigated further before being reported as a finding – specifically, whether it survives when tested only within a single raw source per class rather than across sources.

3. The two-pipeline architecture decision (Section 4 above) should be reviewed and either confirmed or revised, since it is this report's reading of an ambiguous part of the proposal, not something the proposal stated outright.